MTR-QNet: Multi-Task Deep Learning for Regime-Aware Stock Market Analysis

PROJECT REPORT

Submitted By

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CERTIFICATE

This is to certify that the project report entitled ***"MTR-QNet: Multi-Task Deep Learning for Regime-Aware Stock Market Analysis"*** *is a bonafide record of the project work done by* ***Archana J Dev*** *in fulfillment of the requirement for the sixth semester of* ***Bachelor of Computer Application*** *in Department of Computer Science of CHRIST COLLEGE (AUTONOMOUS), IRINJALAKUDA.*

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DECLARATION

I hereby declare that the project entitled **"MTR-QNet: Multi-Task Deep Learning for Regime-Aware Stock Market Analysis"** is a bonafide record of work carried out by me under the guidance of **Mr.ADARSH M D**. This work has not been submitted earlier for the award of any degree or diploma to any university or institution.

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Place: Irinjalakuda

Date:

ACKNOWLEDGEMENT

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ABSTRACT

This report presents the design and implementation of **MTR-QNet: Multi-Task Deep Learning for Regime-Aware Stock Market Analysis**. Stock market forecasting is challenging due to noisy, non-stationary price movements and changing market regimes that traditional models fail to capture. To address this, we propose MTR-QNet, a hybrid multi-task deep learning framework combining dilated Temporal Convolutional Networks (TCNs), GRU-based sequence modeling, and an attention mechanism to learn both short- and long-term patterns from OHLCV data. The model includes an auto-adaptive EMA module with stock-specific parameters and jointly predicts nine next-day market signals, including trend and volatility regimes. These outputs are integrated through an anomaly-aware ensemble scoring system to generate BUY, HOLD, or SELL decisions. Experiments on Indian large-cap stocks show that MTR-QNet outperforms traditional and single-branch deep learning models while remaining lightweight and suitable for real-world trading research.

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Chapter 1

Introduction

1.1 Background

1.1.1 Problem Overview

As financial markets are extremely dynamic, non-linear, and volatile, stock market prediction is a difficult and demanding task. Accurate forecasting is challenging because of the wide range of factors that influence stock prices, including historical trends, trading volume, economic conditions, and investor sentiment. Unreliable predictions can arise from traditional statistical and basic machine learning models' inability to capture abrupt market fluctuations and long-term dependencies. The demand for intelligent systems that can efficiently analyze time-series patterns and produce precise stock price forecasts using cutting-edge deep learning techniques is rising as a result of the expanding availability of large-scale financial data.

1.1.2 Motivation

The need to assist traders, investors, and financial analysts in making well-informed, data-driven decisions is what inspired this project. Predictive systems that can process vast volumes of historical data and derive insightful information are in high demand as financial markets continue to change quickly. Building more reliable forecasting models is made possible by developments in deep learning, especially in the area of recurrent neural networks like Gated Recurrent Units (GRU). The model can concentrate on the most pertinent time steps in the data by incorporating an Attention Mechanism, which increases interpretability and prediction accuracy. In order to make the system accessible to a wider audience, the project also intends to offer an intuitive user interface that makes it simple for users to visualize predictions.

1.1.3 Importance of the Project

In the fields of artificial intelligence and financial analytics, this project is extremely significant. Reducing financial risk, improving decision-making, and optimizing investment strategies are all made possible by accurate stock price prediction. The system shows how sophisticated deep learning techniques, like GRU with Attention Mechanisms, can be used practically in financial forecasting. Additionally, it aids in the creation of sophisticated financial instruments that facilitate automated trading research, portfolio management, and market trend analysis. Overall, the project highlights how AI-driven solutions are becoming more and more important in improving the dependability and efficiency of contemporary financial markets.

1.2 Objectives

1.2.1 Design and Development

The system is designed and developed to provide an intuitive and user-friendly platform for predicting stock market trends. It focuses on creating a structured architecture and smooth interface to ensure ease of use.

1.2.2 Core Functionalities

The project implements key features such as data processing, prediction generation, and result visualization. These functionalities ensure that users can input data and receive accurate and meaningful predictions.

1.2.3 Testing and Validation

The system is thoroughly tested and validated to ensure accuracy, reliability, and consistent performance. Different test scenarios are used to verify that the application functions correctly under various conditions.

1.3 Scope

This project's scope includes designing and developing a deep learning model based on GRU and enhanced with an Attention Mechanism to predict the stock market. In order to produce precise price forecasts, the system focuses on gathering, preprocessing, and evaluating historical stock market data. In order to help users better understand market trends and patterns, it incorporates model training, evaluation, and visualization of prediction results

through an intuitive interface. The project helps investors, analysts, and researchers with market analysis and supports financial decision-making by offering insights into stock price movements. The system does not specifically take into account unforeseen external economic, political, or global events that could affect market behavior because it is mainly based on historical data and predictive modeling.

Chapter 2

Literature Review

2.1 Related Work

The noisy, non-linear nature of financial time-series data frequently causes problems for early machine learning (SVM, Random Forest) and traditional statistical models (ARIMA). Although LSTMs and Recurrent Neural Networks (RNNs) enhanced the modeling of sequential data, they usually fall short in recognizing the significance of particular historical occurrences. A GRU-based architecture combined with an Attention Mechanism is used in this project to overcome these constraints. In a 30-day sequence, this method enables the model to rank the most important time steps first, and the Gated Recurrent Units effectively handle long-term temporal dependencies, producing more reliable and accurate stock market forecasts.

2.2 Technologies Used

- The project uses modern programming tools, deep learning frameworks, and web technologies to build an effective stock market prediction system.
- Python is the main programming language because it excels in data analysis and machine learning.
- Pandas and NumPy libraries handle data preprocessing, cleaning, and manipulation.
- Matplotlib is used for data visualization and model performance analysis.
- The deep learning model is developed with TensorFlow and Keras, which allows the use of GRU layers and the Attention Mechanism.
- Historical stock market data is processed and normalized to improve model accuracy and training efficiency.
- Web technologies like HTML, CSS, and JavaScript are used to create an interactive and user-friendly prediction dashboard.
- These technologies together ensure efficient data handling, accurate model training, and smooth visualization of prediction results.

Chapter 3

System Design

3.1 Overview

Both early machine learning (SVM, Random Forest) and traditional statistical models (ARIMA) frequently have trouble withIn order to increase forecasting accuracy, this project builds a stock market prediction system using a deep learning model based on GRU that is enhanced with an Attention Mechanism. The system predicts future price trends by analyzing historical stock data and offers an interactive user interface that makes it simple to visualize the results. It is a useful instrument for data-driven investment decision-making and financial analysis.

3.2 Architecture

The system architecture follows a client–server model, as illustrated in Figure 5.5, where the frontend and backend work together to create an inter-

active stock market prediction platform. The frontend is built using web technologies such as HTML, CSS, and JavaScript, providing a responsive and user-friendly interface through which users can input stock details and view prediction results. The backend is developed using Python and handles core operations including data processing, execution of the machine learning model, and generation of predictions. The trained GRU with Attention model operates on the backend, processing requests received from the frontend, performing computations, and returning prediction outputs in real time. Communication between the frontend and backend occurs through API requests, ensuring efficient data transfer and quick response handling. This architecture supports modular development, improves scalability, and enables seamless integration between the user interface and the machine learning engine.

3.3 Module Description

- Home Page:

The Home Page is the main entry point of the application. It gives an overview of the stock market prediction system. It introduces users to the platform's purpose and highlights key features like AI-based prediction and analytics. It also offers navigation to other sections, such as login and the dashboard. The page is designed to be visually appealing and user-friendly. This ensures a smooth first interaction with the system.

- Login page:

The Login Page lets users access the system securely by entering their valid credentials. It checks user identity through backend authentication services. This step ensures only authorized users can reach the prediction and dashboard features. This page improves system security and manages user sessions effectively.

- Stock Prediction Page:

The Stock Prediction Page allows users to enter stock-related details and ask for price predictions. User input goes to the backend, where the trained GRU with Attention model processes the information and produces future stock price forecasts. The results appear in both graphical and numerical formats, making the predictions easy to understand.

- AI Chat:

The AI Chat Page lets users engage with a smart assistant for market questions, predictions, and insights. The system keeps chat logs and gives context-based responses to improve user interaction and support decisions. This feature makes it easier to use by providing help through conversational AI.

- About page:

This outlines the platform's mission to make AI-driven stock forecasting available to everyone. It highlights institutional-grade performance metrics, such as accuracy and a user base of over 50,000, to build trust. By detailing the integration of 24/7 AI support and deep learning, the module emphasizes giving retail investors easy access to professional-level market insights.

Chapter 4

Implementation

4.1 Development Tools

The frontend of the system uses HTML, CSS, and JavaScript to create a responsive and easy-to-use interface for input and result visualization. The backend is built with Python, where the machine learning model created with TensorFlow and Keras processes data and generates stock price predictions. API-based communication allows smooth interaction between the frontend and backend, ensuring efficient data flow and real-time prediction delivery.

4.2 Coding Standards

The project follows standard coding practices to ensure clarity, maintainability, and efficiency. It uses meaningful variable and function names, proper indentation, and consistent formatting to improve readability. Modular programming principles organize the code into logical components. Comments

and documentation explain important logic. Error handling techniques and Python best practices are used to ensure reliable, high-quality code.

4.3 Implementation Details

The stock market prediction system has several stages, including data preprocessing, model development, and system integration. First, historical stock data is cleaned, normalized, and turned into time-series sequences for training. A GRU-based deep learning model with an Attention Mechanism is created using TensorFlow and Keras to improve prediction accuracy. The trained model runs on the backend using Python to handle user inputs. Meanwhile, the frontend is built with HTML, CSS, and JavaScript to show prediction results. API communication links the frontend and backend, allowing for real-time predictions and a seamless user experience.

4.3.1 Methodology

- Data Preparation: Uses five years of daily OHLCV data from Indian large-cap stocks. It applies a chronological 80/20 train-test split to avoid look-ahead bias.
- Auto-Adaptive EMA Engine: Dynamically picks the best fast-slow EMA pairs for each stock. It scores their effectiveness in defining bullish and bearish trends, along with trend crossovers.
- Multi-Task Feature Engineering: Creates shifted technical indicators (RSI, MACD, Bollinger Bands) to predict 9 classification labels and 1 regression target (ATR) at the same time.

- Hybrid Backbone Architecture: Uses a dual-branch system with GRU and Attention for long-term context and Dilated TCN for local temporal patterns.

- Ensemble Decision Logic: Combines multi-task predictions with an Isolation Forest anomaly detector to calculate a weighted "Buy Score" for final Buy, Hold, or Sell signals.

4.3.2 Algorithm

MTR-QNet Regime-Aware Multi-Task Pipeline

Step 1: Data Ingestion

Load chronological OHLCV time-series X_t.

Step 2: Normalization and Split

Apply Min–Max scaling and perform chronological 80:20 split.

Step 3: Indicator Engineering

Compute RSI(14), MACD(12,26,9), ATR(14), EMA_{fast}, EMA_{slow}. Shift all indicators by $(t-1)$ to eliminate look-ahead bias.

Step 4: Sequence Encoding

Generate sliding windows ($L = 30$). Define multi-task outputs: 9 classification heads and 1 ATR regression head.

Step 5: Hybrid Feature Extraction

Apply GRU with Attention to obtain F_{seq}. Apply dilated causal TCN to obtain F_{conv}. Fuse embeddings: $Z = [F_{seq} \| F_{conv}]$.

Step 6: Multi-Task Prediction

Apply Softmax for classification and Linear activation for ATR regression.

Step 7: Regime Filtering

Use Isolation Forest on Z to compute anomaly score A_s. Suppress predictions if $A_s > \tau$.

Step 8: Composite Scoring

$$BuyScore = \sum_{i=1}^{9} w_i C_i$$

Step 9: Decision Logic

If $BuyScore \geq \theta_b \rightarrow$ BUY.

If $BuyScore \leq \theta_s \rightarrow$ SELL.

Else HOLD.

Step 10: Volatility Adjustment

$$Upper = Close + k \cdot ATR_{pred}$$

$$Lower = Close - k \cdot ATR_{pred}$$

Step 11: Output

Return {Signal, BuyScore, ATR_{pred}, Upper, Lower}.

4.4 Challenges Faced

- Dealing with noisy and highly unstable stock market data made prediction difficult.
- Adjusting hyperparameters to improve model accuracy needed a lot of testing and time.

- Working with large time-series datasets created challenges in preprocessing and memory management.

- Combining deep learning model outputs with visualization tools required careful timing.

Chapter 5

Testing

5.1 Testing Strategy

The testing strategy for this project guarantees the correctness, dependability, and efficiency of the stock market prediction system. We evaluate the deep learning model using historical stock market data. Its performance is measured with standard metrics like Mean Absolute Error (MAE) and Root Mean Square Error (RMSE). We test individual system components, such as data preprocessing, feature extraction, model training, and prediction generation, to confirm they work correctly. We conduct integration testing to ensure smooth communication between the prediction model and the user interface. Furthermore, we perform validation testing to check that the system provides consistent, accurate, and meaningful results under various input conditions, which helps maintain overall system stability and effectiveness.

5.1.1 Integration Testing:

Integration testing is performed to ensure that different modules of the system work together seamlessly. It verifies the communication between the frontend interface, backend processing, and the prediction model through API interactions. This testing confirms that user inputs are correctly transmitted, processed by the model, and the prediction results are accurately returned and displayed, ensuring smooth end-to-end system functionality.

5.2 Test Cases

1. Test Case 1: Verify that the system loads and preprocesses historical stock market data successfully.

   Expected Result: Data is cleaned, normalized, and prepared correctly without errors.

2. Test Case 2: Verify that the GRU with Attention model trains successfully on the dataset.

   Expected Result: The model finishes training and shows a reduction in loss over epochs.

3. Test Case 3: Verify that the system generates accurate stock price predictions.

   Expected Result: Predicted stock prices closely match actual values within acceptable error limits.

4. Test Case 4: Verify that the system handles invalid or missing input data well.

   Expected Result: The system shows appropriate error messages and continues to work without crashing.

5. Test Case 5: Verify that the user interface displays prediction results and graphs correctly.

   Expected Result: Prediction outputs and visual charts are shown clearly and accurately.

5.3 Results

The completion of all defined test cases confirms that the stock market prediction system functions reliably and meets its design objectives. The GRU with Attention + TCN hybrid model successfully preprocesses historical data, learns time-series patterns, and generates multi-task market predictions. Performance evaluation shows that the Trend Regime classifier achieved 75% accuracy (F1: 0.791), Range Bucket prediction 72.5% (F1: 0.7286), Gap Direction 67.92% (F1: 0.6362), Volume Spike detection 62.08% (F1: 0.7276), Momentum Continuance 54.17%, and Breakout detection a high 92.08% accuracy (F1: 0.8829), while Volatility Regime (47.08%) and EMA-based indicators (30–44.58%) indicate scope for further tuning. The ATR regression model produced an RMSE of 3.8329 with $R^2 = 0.01$, reflecting the difficulty in accurately predicting volatility magnitude. Baseline comparisons showed Logistic Regression (91.67%), Random Forest (92.92%), GRU-only (93.75%), and TCN-only (84.17%) precision, confirming the strength of the

hybrid model in multi-signal prediction despite strong baseline trend classification. The system also detected 21 anomalies and generated stable trading signals.

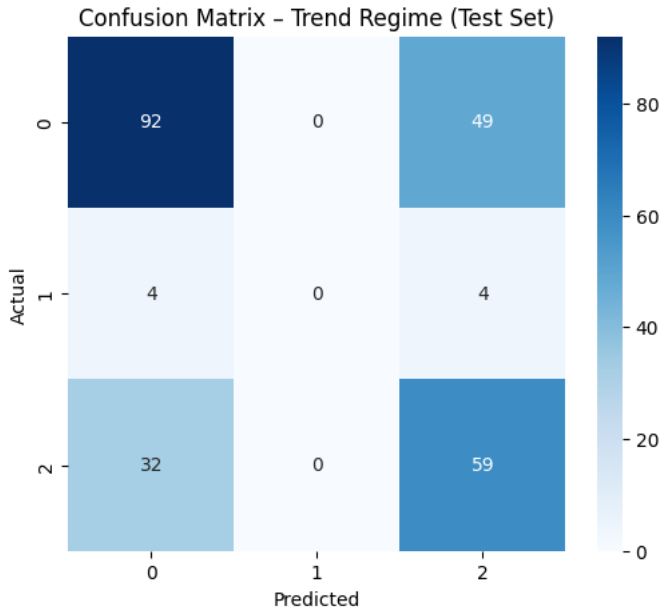


Figure 5.1: Confusion matrix for the trend regime classifier

Figure 5.1 presents the confusion matrix for the trend regime classifier on the test dataset. The model accurately predicts most class 0 and class 2 instances, indicating effective detection of stable and trending market conditions. However, it fails to identify class 1 and shows some confusion between classes 0 and 2, suggesting difficulty in recognizing transitional regimes.

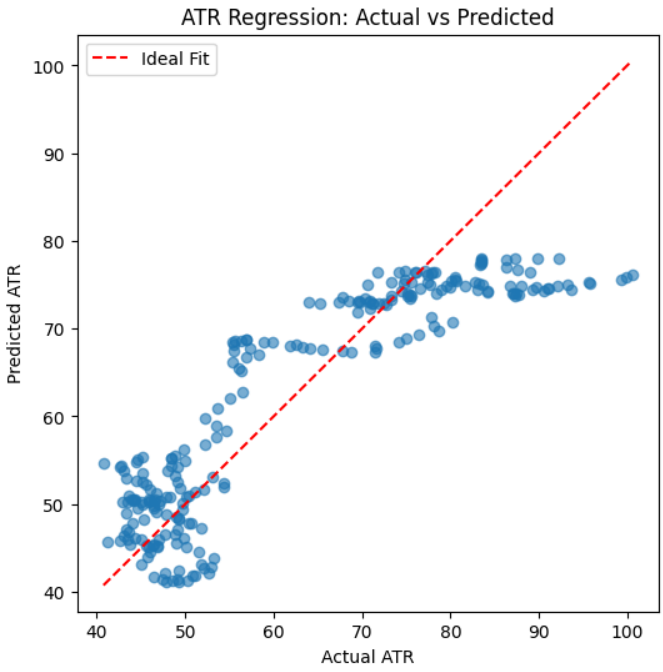


Figure 5.2: ATR regression performance

Figure 5.2 shows the relationship between actual ATR values and the model's predicted ATR. The points generally follow the ideal fit line, indicating that the model captures volatility trends reasonably well. However, some deviations from the line suggest minor prediction errors, particularly at higher ATR values.

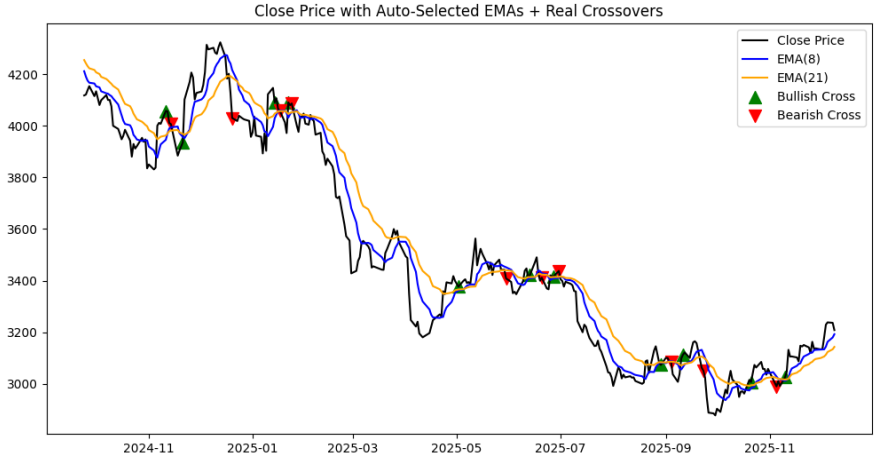


Figure 5.3: Close price with auto-selected EMA pair and crossover markers

Figure 5.3 illustrates the stock's closing price along with auto-selected EMA(8) and EMA(21), highlighting bullish and bearish crossover points. These crossovers indicate potential trend reversals and help identify entry and exit opportunities.

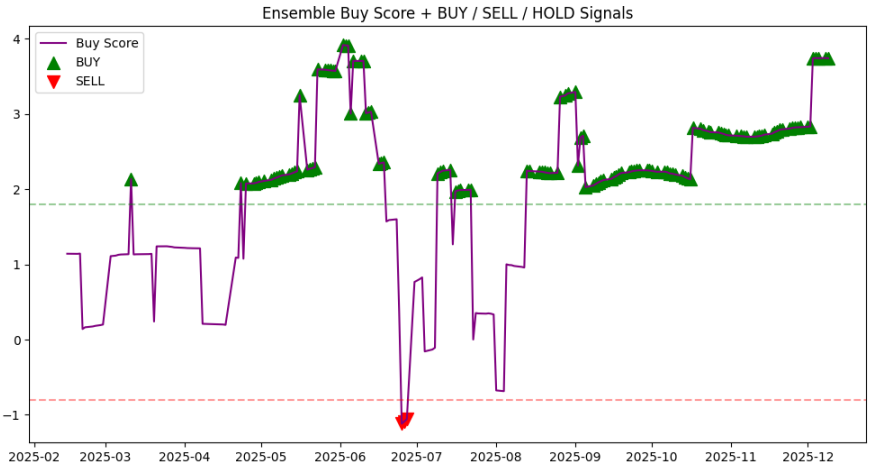


Figure 5.4: Ensemble buy score with BUY and SELL markers

Figure 5.4 presents the ensemble Buy Score used to generate BUY, SELL, and HOLD signals. Scores above the upper threshold indicate BUY signals, while drops below the lower threshold trigger SELL signals, supporting informed trading decisions.

Chapter 6

Conclusion and Future Work

6.1 Conclusion

This project developed a stock market prediction system using a GRU-based deep learning model combined with an Attention Mechanism to improve forecasting accuracy. By analyzing historical stock market data, the system identifies meaningful patterns and predicts future price trends. The implementation shows how machine learning techniques can be used in financial forecasting, while the interactive interface makes it easier to use and understand the results. Overall, the project offers a reliable and smart solution for stock market analysis and supports data-driven investment decisions.

6.2 Future Enhancements

The system can be improved by integrating real-time stock market data to boost prediction accuracy and speed. Deep learning models like Transformers

or hybrid models can be used for better forecasting performance. We can also include features like sentiment analysis from financial news and social media to capture market emotions and trends more effectively. The system can expand into a web or mobile application to enhance accessibility, and we can add automated trading functionalities to support smart, real-time investment decisions.

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APPENDIX

A. Diagrams

A.1 Architecture

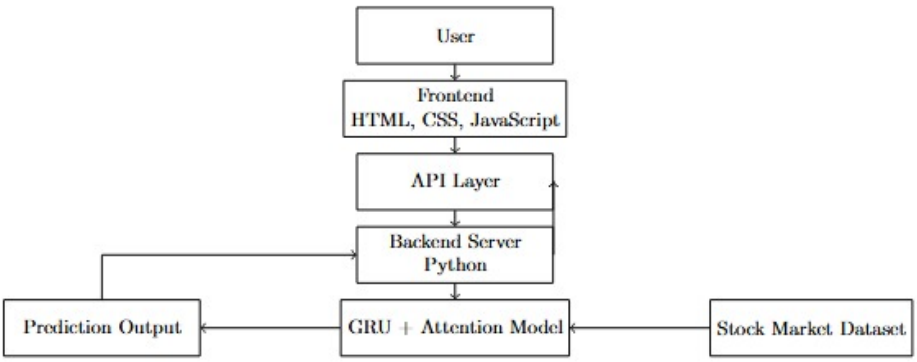


Figure 5.5: Client–Server Architecture of Stock Market Prediction Platform

A.2 Activity Diagram

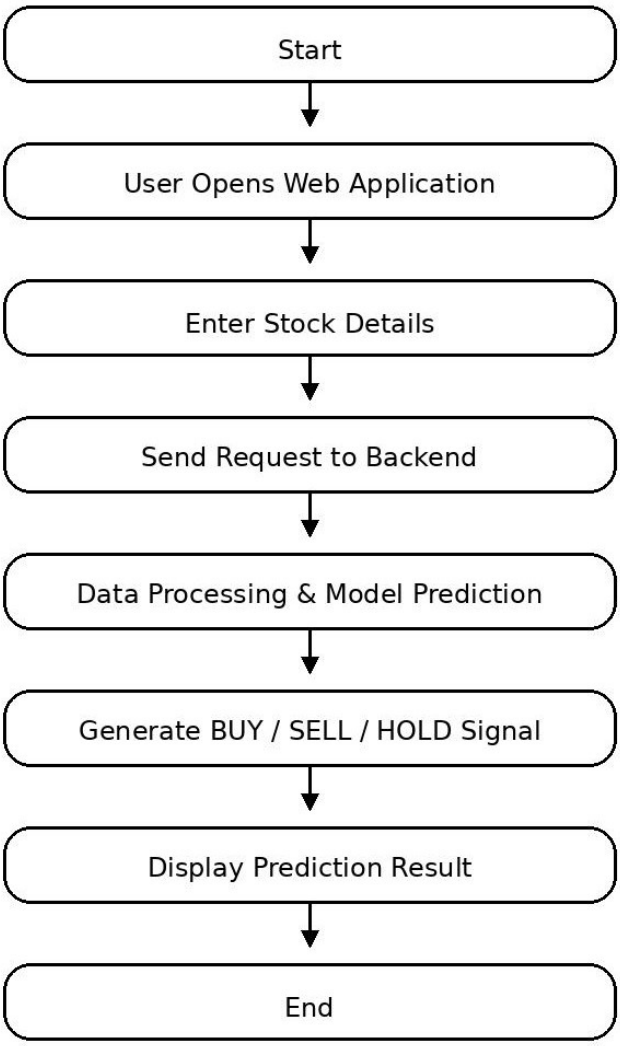


Figure 5.6: Activity Diagram

B. User Interface

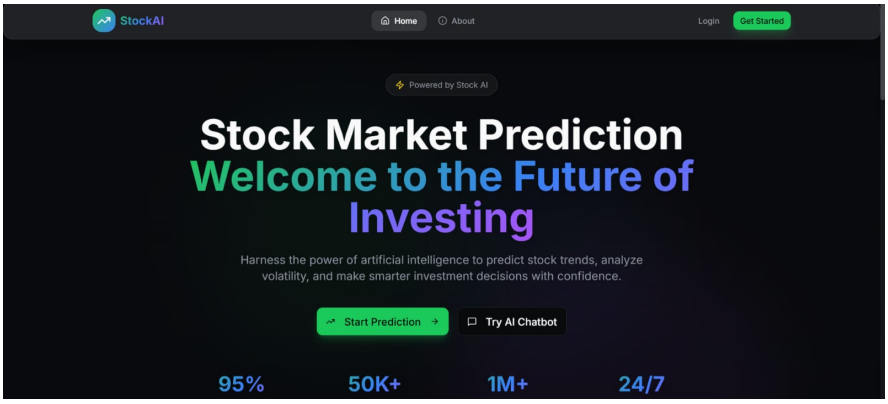


Figure B.1: Home Page

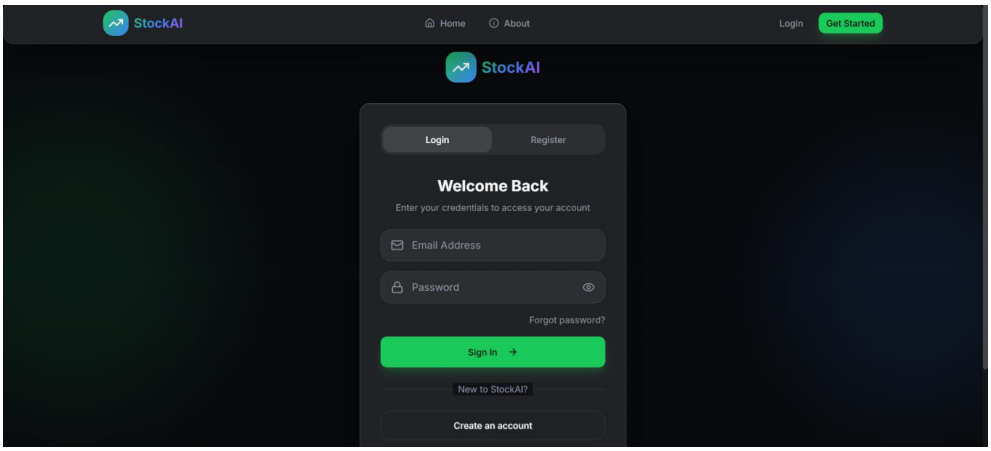


Figure B.2: Login Page

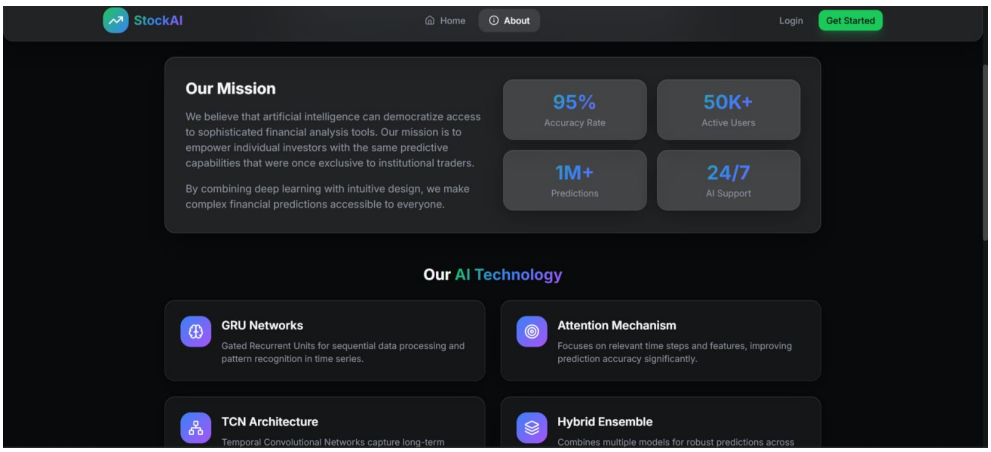


Figure B.3: About Page

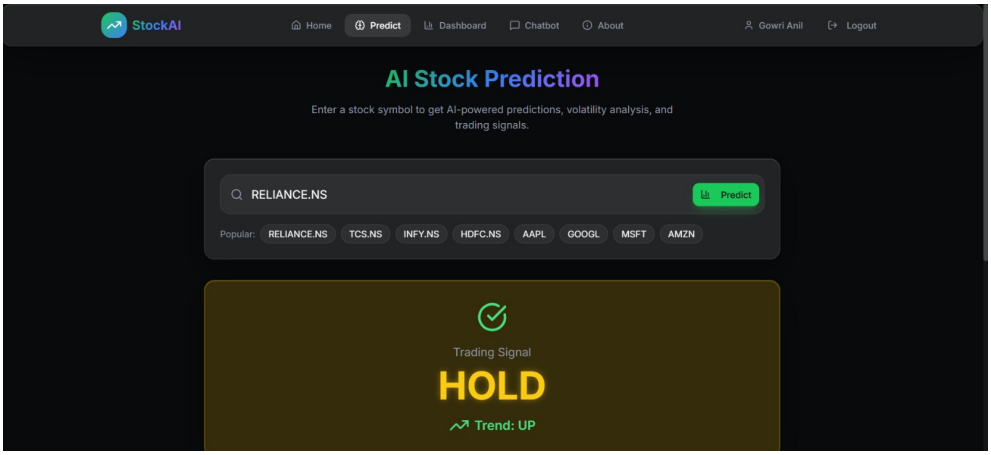


Figure B.4: Stock Page

C. Code

C.1 Data Preprocessing

```
!pip install yfinance ta

# IMPORTS

import numpy as np
import pandas as pd
import yfinance as yf
from ta.trend import EMAIndicator, MACD
from ta.momentum import RSIIndicator
from ta.volatility import BollingerBands, AverageTrueRange


# PARAMETERS

TICKER = "RELIANCE.NS"
```

```
PERIOD = "5y"
INTERVAL = "1d"
RANGE_BINS = [0, 0.005, 0.01, 1]
BREAKOUT_WINDOW = 20
VOLUME_SPIKE_FACTOR = 2.0
CANDIDATE_EMA_PAIRS = [(10,20),(12,26),(20,50),(8,21)]

# DOWNLOAD DATA

df = yf.download(TICKER, period=PERIOD, interval=INTERVAL)

# Flatten columns if MultiIndex
if isinstance(df.columns, pd.MultiIndex):
    df.columns = df.columns.get_level_values(0)

# Ensure numeric
for col in df.columns:
    df[col] = pd.to_numeric(df[col], errors="coerce")

df.dropna(inplace=True)
df.sort_index(inplace=True)
print("Data loaded:", df.shape)

# AUTO-SELECT EMA PAIR
```

```
def auto_select_ema_pair(df, candidate_pairs):
    best_score = -1
    best_pair = candidate_pairs[0]
    close = df["Close"]

    for fast, slow in candidate_pairs:
        if fast >= slow:
            continue
        ema_fast = EMAIndicator(close, fast).ema_indicator()
        ema_slow = EMAIndicator(close, slow).ema_indicator()
        diff = ema_fast - ema_slow
        zone = np.sign(diff)
        valid = zone[zone != 0]
        if len(valid) < 50:
            continue
        bull_frac = (valid == 1).mean()
        bear_frac = (valid == -1).mean()
        imbalance = abs(bull_frac - bear_frac)
        cross_rate = ((valid * valid.shift(1)) == -1).sum() / len(valid)
        score = (1 - imbalance) + min(cross_rate*10, 1)
        if score > best_score:
            best_score = score
            best_pair = (fast, slow)
    print(f"Selected EMA pair: {best_pair}, score: {best_score:.3f}")
    return best_pair
```

```
EMA_FAST, EMA_SLOW = auto_select_ema_pair(df, CANDIDATE_EMA_PAIRS)

# COMPUTE TECHNICAL INDICATORS

df["EMA_fast"] = EMAIndicator(df["Close"], EMA_FAST).ema_indicator()
df["EMA_slow"] = EMAIndicator(df["Close"], EMA_SLOW).ema_indicator()
df["EMA_diff"] = df["EMA_fast"] - df["EMA_slow"]

macd = MACD(df["Close"])
df["MACD"] = macd.macd()
df["MACD_Signal"] = macd.macd_signal()
df["MACD_Hist"] = macd.macd_diff()

df["RSI"] = RSIIndicator(df["Close"], 14).rsi()
bb = BollingerBands(df["Close"], 20, 2)
df["BB_Width"] = bb.bollinger_hband() - bb.bollinger_lband()
df["ATR_14"] = AverageTrueRange(df["High"], df["Low"], df["Close"],
14).average_true_range()

# NEXT-DAY LABELS

df["Close_Lag1"] = df["Close"].shift(1)
df["Return"] = df["Close"].pct_change()
df["Range_pct_next"] = ((df["High"] - df["Low"]) / df["Close"]).shift(-1)
```

```
df["Range_Bucket_raw"] = pd.cut(df["Range_pct_next"], bins=RANGE_BINS,
labels=[0,1,2])
df["Gap_pct"] = (df["Open"] - df["Close"].shift(1)) / df["Close"].shift(1)
df["Gap_Label_raw"] = pd.cut(df["Gap_pct"], bins=[-1,-0.005,0.005,1],
labels=[-1,0,1])
df["Vol_next"] = df["Volume"].shift(-1)
df["Vol_median20"] = df["Volume"].rolling(20).median()
df["Vol_Spike_raw"] = (df["Vol_next"] > df["Vol_median20"]*VOLUME_SPIKE_FACTOR)
.astype(int)
df["Return_next"] = df["Return"].shift(-1)
df["Momentum_Continue_raw"] = (np.sign(df["Return"]) == np.sign
(df["Return_next"])).astype(int)
df["20d_high"] = df["High"].rolling(BREAKOUT_WINDOW).max()
df["Close_next"] = df["Close"].shift(-1)
df["Breakout_raw"] = (df["Close_next"] > df["20d_high"]).astype(int)

# EMA next day for crossover/zone
df["EMA_fast_next"] = df["EMA_fast"].shift(-1)
df["EMA_slow_next"] = df["EMA_slow"].shift(-1)

def ema_cross_real(row):
    prev = row["EMA_fast"] - row["EMA_slow"]
    nxt = row["EMA_fast_next"] - row["EMA_slow_next"]
    if prev <= 0 and nxt > 0: return 1
    if prev >= 0 and nxt < 0: return -1
```

```
    return 0
df["EMA_Crossover_raw"] = df.apply(ema_cross_real, axis=1)

def ema_zone_next(row):
    if row["EMA_fast_next"] > row["EMA_slow_next"]*1.001: return 1
    if row["EMA_fast_next"] < row["EMA_slow_next"]*0.999: return -1
    return 0
df["EMA_Trend_Zone_raw"] = df.apply(ema_zone_next, axis=1)
df["ATR_next"] = df["ATR_14"].shift(-1)

# LEAK-FREE SHIFT

shift_cols = ["EMA_fast","EMA_slow","EMA_diff","MACD","MACD_Signal","MACD_Hist",
"RSI","BB_Width","ATR_14","Close_Lag1","Return","Vol_median20","20d_high"]
for col in shift_cols:
    df[col] = df[col].shift(1)

df.dropna(inplace=True)

# FINAL LABELS

df["Range_Bucket"] = df["Range_Bucket_raw"].astype("Int64")
df["Gap_Label"] = df["Gap_Label_raw"].astype("Int64")
df["Vol_Spike"] = df["Vol_Spike_raw"]
df["Momentum_Continue"] = df["Momentum_Continue_raw"]
```

```
df["Breakout"] = df["Breakout_raw"]
df["EMA_Crossover"] = df["EMA_Crossover_raw"].astype(int)
df["EMA_Trend_Zone"] = df["EMA_Trend_Zone_raw"].astype(int)
df["ATR_Target"] = df["ATR_next"].astype(float)

print("Preprocessing complete. Dataset shape:", df.shape)
```

C.2 Training

```
# INSTALL REQUIRED PACKAGES

!pip install torch torchmetrics numpy scikit-learn


# IMPORTS

import torch
import torch.nn as nn
from torch.utils.data import DataLoader, TensorDataset, random_split
from sklearn.preprocessing import StandardScaler
import numpy as np
```

```
# DEVICE

device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
print("Using device:", device)



# PREPARE DATA FOR TRAINING

# Select features and label
features = ["EMA_fast","EMA_slow","EMA_diff","MACD","MACD_Signal","MACD_Hist",
"RSI","BB_Width","ATR_14","Close_Lag1","Return","Vol_median20","20d_high"]
label = "Breakout"  # example, can change to other targets

X = df[features].values
y = df[label].values

# Normalize features
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# Convert to PyTorch tensors
X_tensor = torch.tensor(X_scaled, dtype=torch.float32)
y_tensor = torch.tensor(y, dtype=torch.float32).unsqueeze(1)

# Create Dataset and DataLoader
```

```
dataset = TensorDataset(X_tensor, y_tensor)
train_size = int(0.8 * len(dataset))
test_size = len(dataset) - train_size
train_dataset, test_dataset = random_split(dataset, [train_size, test_size])
train_loader = DataLoader(train_dataset, batch_size=32, shuffle=True)
test_loader = DataLoader(test_dataset, batch_size=32, shuffle=False)



# MODEL DEFINITION (GRU + TCN)

class GRU_TCN(nn.Module):
def __init__(self, input_size,
hidden_size=64, tcn_channels=[32,16], output_size=1):
        super(GRU_TCN, self).__init__()
        self.gru = nn.GRU(input_size, hidden_size, batch_first=True)

        # Simple 1D TCN using dilated conv
        layers = []
        in_ch = 1
        for out_ch in tcn_channels:
            layers.append(nn.Conv1d(in_ch, out_ch, kernel_size=2, padding=1,
            dilation=2))
            layers.append(nn.ReLU())
            in_ch = out_ch
        self.tcn = nn.Sequential(*layers)
```

```
        self.fc = nn.Sequential(
            nn.Linear(hidden_size + tcn_channels[-1], 32),
            nn.ReLU(),
            nn.Linear(32, output_size),
            nn.Sigmoid()  # for binary classification
        )

    def forward(self, x):
        # GRU path
        x_gru, _ = self.gru(x.unsqueeze(1))  # batch, seq_len=1, features
        x_gru = x_gru[:, -1, :]

        # TCN path
        x_tcn = x.unsqueeze(1)  # batch, channels=1, features
        x_tcn = self.tcn(x_tcn)
        x_tcn = x_tcn.mean(dim=2)  # global average pooling

        # Concatenate
        x_cat = torch.cat([x_gru, x_tcn], dim=1)
        out = self.fc(x_cat)
        return out

model = GRU_TCN(input_size=X_tensor.shape[1]).to(device)
print(model)
```

```
# LOSS AND OPTIMIZER

criterion = nn.BCELoss()  # binary label
optimizer = torch.optim.Adam(model.parameters(), lr=0.001)



# TRAINING LOOP

epochs = 30
for epoch in range(epochs):
    model.train()
    train_loss = 0
    for xb, yb in train_loader:
        xb, yb = xb.to(device), yb.to(device)
        optimizer.zero_grad()
        y_pred = model(xb)
        loss = criterion(y_pred, yb)
        loss.backward()
        optimizer.step()
        train_loss += loss.item() * xb.size(0)
    train_loss /= len(train_loader.dataset)

    # Validation
```

```
    model.eval()
    val_loss = 0
    with torch.no_grad():
        for xb, yb in test_loader:
            xb, yb = xb.to(device), yb.to(device)
            y_pred = model(xb)
            loss = criterion(y_pred, yb)
            val_loss += loss.item() * xb.size(0)
    val_loss /= len(test_loader.dataset)

 print(f"Epoch {epoch+1}/{epochs} | Train Loss: {train_loss:.4f} |
 Val Loss: {val_loss:.4f}")



# SAVE MODEL

torch.save(model.state_dict(), "gru_tcn_stock_model.pth")
print("Model saved as gru_tcn_stock_model.pth")
```

C.3 Testing

```
# TESTING / INFERENCE
```

```
import torch
from sklearn.metrics import accuracy_score, f1_score, confusion_matrix,
classification_report

# Load the saved model
model = GRU_TCN(input_size=X_tensor.shape[1]).to(device)
model.load_state_dict(torch.load("gru_tcn_stock_model.pth"))
model.eval()
print("Model loaded for testing.")

# Store predictions and true labels
y_true = []
y_pred = []

with torch.no_grad():
    for xb, yb in test_loader:
        xb, yb = xb.to(device), yb.to(device)
        outputs = model(xb)
        predicted = (outputs > 0.5).float()
        # threshold for binary classification
        y_true.extend(yb.cpu().numpy())
        y_pred.extend(predicted.cpu().numpy())

y_true = np.array(y_true)
y_pred = np.array(y_pred)
```

```
# Accuracy
acc = accuracy_score(y_true, y_pred)
f1 = f1_score(y_true, y_pred)
cm = confusion_matrix(y_true, y_pred)

print(f"Test Accuracy: {acc:.4f}")
print(f"F1 Score: {f1:.4f}")
print("Confusion Matrix:")
print(cm)
print("\nClassification Report:")
print(classification_report(y_true, y_pred))



# PREDICTION ON NEW DATA

# Suppose 'new_data' is a DataFrame with same features as training
new_data_scaled = scaler.transform(new_data[features].values)
new_tensor = torch.tensor(new_data_scaled, dtype=torch.float32).to(device)

with torch.no_grad():
    predictions = model(new_tensor)
    predictions = (predictions > 0.5).float().cpu().numpy()

print("Predictions for new data:")
```

```
print(predictions)
```